Example 12.23

A main road in a City has 4 crossroads with traffic lights. Each traffic light opens or closes the traffic with the probability of 0.4 and 0.6 respectively. Determine the probability of

(i) a car crossing the first crossroad without stopping
(ii) a car crossing first two crossroads without stopping
(iii) a car crossing all the crossroads, stopping at third cross.
(iv) a car crossing all the crossroads, stopping at exactly one cross.

Solution:

Let A_i be the event that the traffic light opens at i th cross, for i= 1, 2, 3, 4.

Let B_i be the event that the traffic light closes at i th cross, for i= 1, 2, 3, 4.

The traffic lights are all independent.

Therefore A_i and B_i are all independent events, for i= 1, 2, 3, 4.

Given that

$$P(A_i) = 0.4,\ i = 1, 2, 3, 4$$

$$P(B_i) = 0.6,\ i = 1, 2, 3, 4$$

(i) Probability of car crossing the first crossroad without stopping,

$$P(A_1) = 0.4.$$

(ii) Probability of car crossing first two crossroads without stopping,

$$P(A_1 \cap A_2) = P(A_1A_2) = (0.4)(0.4) = 0.16$$

(iii) Probability of car crossing all the crossroads, stopping at third cross

$$P(A_1 \cap A_2 \cap B_3 \cap A_4) = P(A_1A_2B_3A_4) = (0.4)(0.4)(0.6)(0.4) = 0.0384$$

(iv) Probability of car crossing all the crossroads, stopping at exactly one of the crossroads is

$$P(B_1A_2A_3A_4 \cup A_1B_2A_3A_4 \cup A_1A_2B_3A_4 \cup A_1A_2A_3B_4)$$

$$= P(B_1A_2A_3A_4) + P(A_1B_2A_3A_4) + P(A_1A_2B_3A_4) + P(A_1A_2A_3B_4)$$

$$= 4(0.4)(0.4)(0.6)(0.4) = 4(0.0384) = 0.1536$$

EXERCISE 12.3

(1) Can two events be mutually exclusive and independent simultaneously?

(2) If A and B are two events such that $P(A \cup B) = 0.7$, $P(A \cap B) = 0.2$, and $P(B) = 0.5$, then show that A and B are independent.

(3) If A and B are two independent events such that $P(A \cup B) = 0.6$, $P(A) = 0.2$, find $P(B)$.

(4) If $P(A) = 0.5$, $P(B) = 0.8$ and $P(B/A) = 0.8$, find $P(A / B)$ and $P(A \cup B)$.

(5) If for two events A and B, $P(A) = \frac{3}{4}$, $P(B) = \frac{2}{5}$ and $A \cup B = S$ (sample space), find the conditional probability $P(A / B)$.

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